MARKO TOT

Game AI Researcher | PhD Student

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From the youngest age, I have been interested in video games and solving problems. Now, as a PhD. student in computer science, specializing in artificial intelligence I can give my personal contribution to the AI and have my childhood dreams become a reality.

PROGRAMMING LANGUAGES

- Python
- C
- C++
- C#
- Java

FRAMEWORKS

- PyTorch
- Tensorflow
- NetworkX
- Unity

SKILLS

- Model Based RL
- Statistical Forward Planning
- Object Oriented Programming
- Game AI algorithms

LANGUAGES

- **English** *(full professional proficiency)*
- **Serbian** *(native proficiency)*
- **German** (elementary proficiency)

WORK EXPERIENCE

Student Demonstrator | Queen Mary University of London

10/2021 – 02/2022 — *London, United Kingdom*

- Taught AI in Games course that covered planning algorithms and deep reinforcement learning.
- Taught Multiplatform Game Development course that covered all aspects of game development.

Teaching Assistant | Faculty of Technical Sciences

09/2016 – 09/2020 — *Novi Sad, Serbia*

- Organized lectures, lab exercises and assignments for several computer science courses including: Data Structures (C), Object Oriented Programming (C++, Java), Electric Power Software Development (C#), and Human Computer Interaction. (C#)
- Lead the Faculty of Technical Sciences team for the international university competition in Object Oriented Programming. (C++)

Research Software Developer | Horizon 2020

10/2018 – 09/2019 — *Novi Sad, Serbia*

- Adapted the PhysiCell framework for artificial evolution and validation of novel strategies for cancer treatment using nanoparticles for EVO-NANO project.
- Created a new AI generated vascular subsystem and integrated in with an established large scale cell simulator. (C++)

Junior Game Developer | Panza Games

06/2016 – 11/2016 — *Novi Sad, Serbia*

- As a game developer in Panza Games I created several 2D games for PC and Android.
- I was tasked with creating gameplay mechanics, AI behaviour, netcode and multiplayer synchronization. (Unity)

EDUCATION

PhD student | Queen Mary University of London

09/2020 – PRESENT — *London, United Kingdom*

- PhD Topic: Local Forward Model Learning for Statistical Forward Planning
- My current research is focused on creating a learned model based on combining symbolic features and local surroundings to give accurate multi-step predictions. (NetworkX, PyTorch)
- I have received the Microsoft Research PhD Scholarship in 2020 and am a part of IGGI CDT and GameAI research group in Queen Mary University of London.

PUBLICATIONS

What Are You Looking At? Team Fight Prediction Through Player Camera

Conference on Games, 2021

The study showcases a model that can predict and detect team fights in the game of Dota 2. The approach outlines a novel technique of deriving state representation of a complex game environment by relying on player's knowledge. This is done through analysing the position of the in-game characters and their associated cameras and utilizing this data to train a neural network.
The full paper is available here:
http://diego-perez.net/papers/TeamFightPredictionCoG2021.pdf

Making Something Out of Nothing: Monte Carlo Graph Search in Sparse Reward Environments

AAAI-22 Workshop on RL and Games, 2022

The study improves the sample efficiency of planning-based algorithm called Monte Carlo Graph Search in sparse reward environments through utilizing a frontier, stored rollouts and novelty metrics.
The full paper is available here:
http://diego-perez.net/papers/MCGS-AAAIWorkshop22.pdf

Turning Zeroes into Non-Zeroes: Sample Efficient Exploration with Monte Carlo Graph Search

Conference on Games, 2022

The study adapts and optimizes the performance Monte Carlo Graph Search extensions such as frontier, stored rollouts and novelty metrics to stochastic sparse reward environments.
The full paper is available here:
http://diego-perez.net/papers/TurningZeroesMCGS-CoG22.pdf

PERSONAL PROJECTS

Runners	Runners is a 2D game that incorporated real time **multi-agent collaborative pathfinding algorithm**. (Java)
DinosHur	DinosHur is a 3D racing/shooting game with **pathfinding**, **AI steering and dynamic rubber-band mechanism**. (Unity)
Jump & Squat	Jump and Squat is a real-time motion tracking game. It uses camera input and works based on **object detection and motion tracking**. (Python, MediaPipe, OpenCV)
Simple Life	Simple Life is a 2D Grid Based, partially observable games in which an agent needs to find and collect food to survive. The agent is a RL based agents implemented through a **recurrent DQN network**. (TensorFlow)
Snake AI	Snake AI is a 2-player competitive snake game. The agents were implemented and trained through a **self-play system**. (Unity, ML-Agents)